CASE STUDY 2: Traffic Flow and Velocity Estimation

Scenario:

A traffic monitoring system records the position of a vehicle every second. Engineers want to compute velocity and

total distance traveled.

Computational Task

Students must:

• Estimate instantaneous velocity

• Compute total distance

• Analyze traffic behavior

Given Data:

TIME (s)	POSITION(m)
0	0
1	5
2	15
3	30
4	50
5	75

1. Velocity Estimation

• Use central difference:

$$\frac{v(t) = x(t+1) - x(t-1)}{2}$$

Compute velocity at:

t = 1, 2, 3, 4

- **t = 1**

$$v(1) = \frac{15 - 0}{2} = 7.5$$

- **t = 2**

$$v(2) = \frac{30 - 5}{2} = 12.5$$

- **t = 3**

$$v(3) = \frac{50 - 15}{2} = 17.5$$

- **t = 4**

$$v(4) = \frac{75 - 30}{2} = 22.5$$

Time (s)	Velocity (m/s)
1	7.5
2	12.5
3	17.5
4	22.5

2. ACCELERATION INSIGHT

To determine acceleration, we apply numerical differentiation again to velocity using the **central difference formula**:

$$a(t) = \frac{v(t+1) - v(t-1)}{2h}$$

Since the time interval is **1 second**, then:

$$h = 1 \Rightarrow a(t) = \frac{v(t+1) - v(t-1)}{2}$$

At t = 2

$$a(2) = \frac{v(3) - v(1)}{2} = \frac{17.5 - 7.5}{2} = 5 \text{ m/s}^2$$

At t = 3

$$a(3) = \frac{v(4) - v(2)}{2} = \frac{22.5 - 12.5}{2} = 5 \text{ m/s}^2$$

Time (s)	Acceleration (m/s²)
2	5
3	5

3. Distance Verification

- Use Trapezoidal Rule to compute:

$$\int v(t)\,dt$$

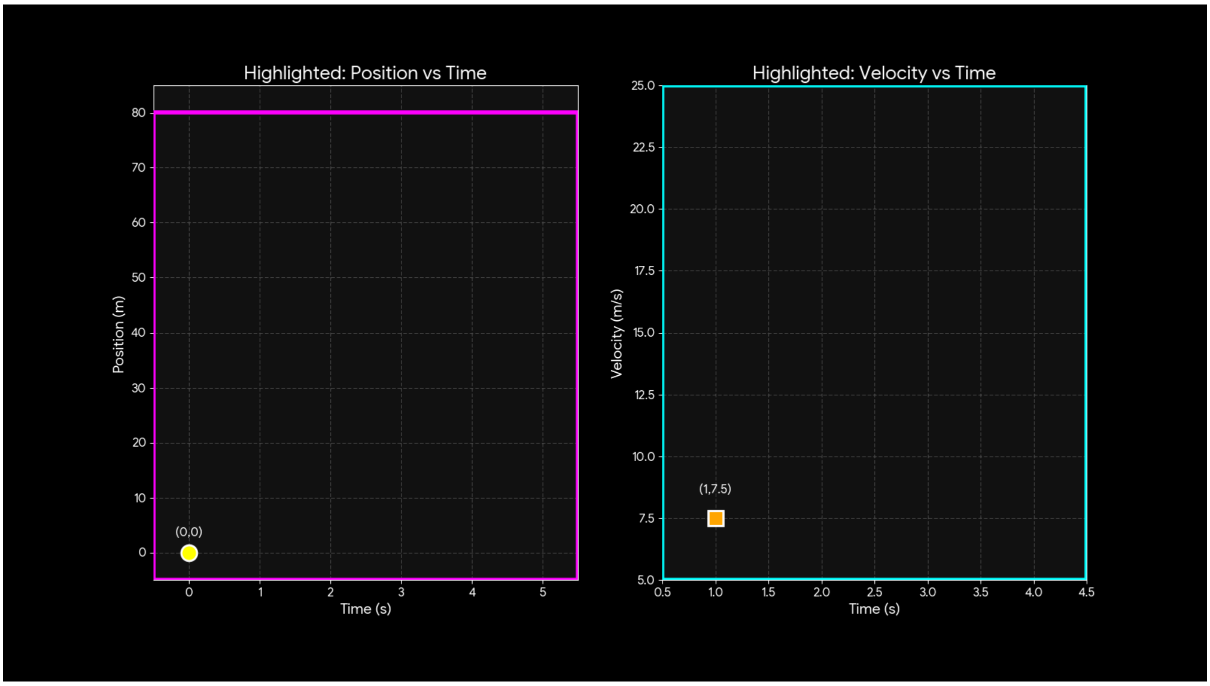
We integrate velocity:

$$\int v(t)\,dt \approx \frac{h}{2}[v_1 + 2v_2 + 2v_3 + v_4]$$

Where **h = 1 second**

$$= \frac{1}{2}[7.5 + 2(12.5) + 2(17.5) + 22.5]$$
$$= \frac{1}{2}[7.5 + 25 + 35 + 22.5]$$
$$= \frac{1}{2}(90) = 45 \text{ meters}$$

4. Visual Representation



5. Analysis Questions

• When is the car accelerating?

- Entire duration (velocity keeps increasing)

• Is motion uniform?

- No (velocity changes every second)

• Identify any anomalies (sudden jumps)

- None, data is consistent

Summary:

This case study examines the movement of a vehicle by analyzing its **position**, **velocity**, and **acceleration** over a 5-second interval. Here is a short breakdown of the computational tasks performed:

1. Estimating Velocity

Engineers used a **central difference formula** to estimate how fast the car was moving at specific points in time. By looking at the position of the car one second before and one second after a specific time, they determined that the velocity was steadily increasing from **7.5 m/s** at t = 1 to **22.5 m/s** at t = 4.

2. Analyzing Acceleration

To find the acceleration, the same numerical method was applied to the calculated velocities.

- **Constant Rate**: The calculations at t = 2 and t = 3 both resulted in an acceleration of **5 m/s²**.
- **Insight**: This indicates that the vehicle was speeding up at a consistent, uniform rate throughout that period.

3. Calculating Distance

The **Trapezoidal Rule** was used to verify the total distance traveled by integrating the velocity over time.

- **Formula**: The integration used the formula $\int v(t)dt \approx \frac{h}{2}\ [v_1 +\ 2v_2 +\ 2v_3 + v_4]$
- **Result**: Based on the velocities from seconds 1 to 4, the calculated distance traveled during that specific window was **45 meters**.

4. Key Findings

- **Non-Uniform Motion**: The motion is not uniform because the velocity changes every second.
- **Consistency**: The data shows a smooth increase in speed with no sudden jumps or anomalies, suggesting the vehicle was accelerating steadily for the entire duration.